

Assignment 03: Case-based Applications
MGT 3202: Engineering Management
Summer 2025-2026

Total Marks: 20; Submission Deadline: Saturday, 19th September 2026

Answer all the following questions:

Question 01: State whether the following statements are True or False. If a statement is False, write the correct statement.

- a) Customer demand information may move upstream in a supply chain even though physical products generally move toward customers.
- b) Locating several distribution facilities close to customers may improve responsiveness, although it can increase facility related costs.
- c) Under a perpetual inventory system, inventory records are updated continuously as items enter or leave stock.
- d) Safety stock is mainly maintained to satisfy demand that is known with certainty in advance.
- e) Under the basic EOQ model, increasing order quantity increases average inventory but normally reduces the number of orders placed during a year.
- f) EOQ mainly answers how much to order, while the reorder point mainly answers when to place the next order.
- g) When total demand is greater than total supply in a transportation problem, a dummy source may be added to balance the problem.
- h) A Basic Feasible Solution obtained using the Least Cost Method is always guaranteed to provide the minimum possible transportation cost.
- i) An activity with zero slack cannot be delayed without affecting the scheduled completion time of the project.
- j) Reducing the duration of any noncritical activity will automatically reduce the overall project duration.

Question 02: Select the most appropriate answer.

- a) A supplier sends an invoice to a buyer and the buyer later makes payment according to the agreed credit terms. This mainly represents:
 - a. Product flow
 - b. Information flow
 - c. Financial flow
 - d. Transportation flow
- b) A manufacturer decides to purchase a component from a specialist supplier instead of producing it internally. Which supply chain driver is mainly involved?
 - a. Inventory
 - b. Sourcing
 - c. Pricing
 - d. Facilities

- c) Total supply in a transportation problem is 600 units, while total demand is 540 units. To balance the problem, which of the following is required?
- Dummy source of 60 units
 - Dummy destination of 60 units
 - Dummy source of 540 units
 - No dummy is required
- d) A retailer checks its inventory every Sunday and decides how much to order based on the stock available at that time. Which inventory system is being used?
- Perpetual review system
 - Continuous replenishment system
 - Periodic review system
 - Pipeline inventory system
- e) An item costs Tk. 500 and its annual holding cost is estimated at 20 percent of the item value. What is the annual holding cost per unit?
- Tk. 20
 - Tk. 50
 - Tk. 100
 - Tk. 400
- f) If all other factors remain unchanged, an increase in ordering cost will generally cause EOQ to:
- Increase
 - Decrease
 - Become zero
 - Remain unchanged
- g) Average daily demand is 40 units and supplier lead time is 5 days. Assuming demand and lead time are constant, the reorder point is:
- 8 units
 - 40 units
 - 200 units
 - 225 units
- h) During which stage of a project does management mainly compare actual progress with planned progress and take corrective action when required?
- Initiating
 - Planning
 - Monitoring and controlling
 - Closing
- i) During backward calculation, if an activity has several immediate successors, its latest finish time is determined from the:
- Highest ES among the successors
 - Lowest LS among the successors
 - Highest EF among the predecessors

- d. Average LS among the successors
- j) Which project management tool is useful for showing when activities are planned to start and finish and for comparing planned progress with actual progress?
 - a. EOQ model
 - b. Gantt chart
 - c. Transportation matrix
 - d. Break even chart

Question 03:

For Questions a to d, identify the main supply chain driver being used. Also state whether the decision mainly supports responsiveness or efficiency. Briefly explain your answer.

- a) Rapid Grocer Ltd. has opened several small fulfillment centers near densely populated areas so that online grocery orders can reach customers within one hour. Operating several facilities increases cost, but delivery time is reduced considerably.
- b) Pharma Track Ltd. uses real time sales, inventory, and shipment information from all of its depots. If one depot is expected to face a stockout, inventory can be moved from another location before customers are affected.
- c) Budget Mart Ltd. keeps a limited range of fast moving products and maintains relatively low inventory levels. This helps the company reduce storage, handling, and obsolescence costs.
- d) Tech Care Ltd. uses specialist third party companies for product repairs and reverse logistics instead of developing these activities internally. This allows the company to avoid maintaining dedicated internal resources and reduce the cost of these activities.

Answer Questions e to h briefly.

- e) A company manufactures the same household appliance every working day. The same company is also installing a completely new robotic assembly line with a fixed budget, defined deliverables, and a six month completion target. Identify which activity is an operation and which is a project. Explain briefly.
- f) A pharmacy reviews its inventory every Friday and then decides how much medicine should be ordered to meet expected demand until the next review. Which inventory review system is being used? Explain briefly.
- g) A transportation problem has total supply of 340 units and total destination demand of 365 units. What adjustment should be made before applying LCM? What does this adjustment represent?
- h) A project manager wants a simple visual schedule showing when each activity is expected to start and finish. The manager also wants to compare actual progress with the original schedule later. Which project management tool would be most suitable?

Question 04: Fresh Foods Distribution Ltd. supplies packaged food products from three warehouses to three regional markets. The transportation cost per carton, available supply, and market demand are given below.

Source	Dhaka	Rajshahi	Khulna	Supply
Gazipur	5	8	3	70
Narayanganj	6	2	9	60
Cumilla	4	7	1	50
Demand	80	55	45	

- Determine whether the transportation problem is balanced.
- Using the Least Cost Method, determine a Basic Feasible Solution.
- Calculate the total transportation cost.

Question 05: Medi Care Supplies Ltd. distributes medical consumables from four depots to three hospitals. The transportation cost per unit, available supply, and hospital demand are given below.

Source	Hospital A	Hospital B	Hospital C	Supply
Depot 1	9	4	11	55
Depot 2	6	12	3	45
Depot 3	5	8	10	60
Depot 4	7	2	13	40
Demand	80	70	65	

- Determine whether the transportation problem is balanced.
- If necessary, add the appropriate dummy source or destination.
- Determine a Basic Feasible Solution using LCM. Show the allocation sequence clearly.
- Calculate the total transportation cost based on your allocation.
- Explain what the dummy allocation represents in this situation.

Question 06: Home Tech Appliances Ltd. transports household appliances from three factories to four distribution centers.

Source	North DC	South DC	East DC	West DC	Supply
Factory A	7	3	11	5	90
Factory B	9	6	2	12	75
Factory C	4	10	8	1	65
Demand	55	60	50	45	

- Determine the difference between total supply and total demand.
- Balance the transportation problem appropriately.
- Apply the Least Cost Method and determine a Basic Feasible Solution. Show the allocation sequence clearly.
- Calculate the total transportation cost based on your allocation.
- Explain what the dummy destination represents in this case.

Question 07: Agro Fresh Logistics Ltd. transports processed agricultural products from four collection centers to four wholesale markets.

Source	Market 1	Market 2	Market 3	Market 4	Supply
Center A	8	3	11	14	75
Center B	5	12	2	9	65
Center C	15	6	10	1	55
Center D	4	13	7	16	45
Demand	60	50	70	60	

- Check whether the transportation problem is balanced.
- Determine the allocation using LCM.
- Calculate the total transportation cost.
- Identify the route receiving the largest allocation in your solution.

Question 08: Nova Consumer Products Ltd. operates five production centers and supplies five regional distribution hubs. The transportation cost per unit, production capacity, and hub demand are given below.

Source	Hub A	Hub B	Hub C	Hub D	Hub E	Supply
Plant 1	14	7	18	4	22	120
Plant 2	9	16	3	20	11	100
Plant 3	5	24	13	8	19	90
Plant 4	17	2	21	10	15	80
Plant 5	6	12	25	1	23	70
Demand	85	95	100	90	70	

- Determine whether the transportation problem is balanced.
- Make the required adjustment to balance the problem.
- Determine the transportation plan using LCM. Show the allocation sequence clearly.
- Calculate the total transportation cost based on your allocation.
- If unused production capacity appears in the final allocation, explain what it means from an operational point of view.

Question 09: OfficePrint Ltd. sells an average of 2,400 toner cartridges per month. The company operates 288 working days per year. Its supplier normally takes 6 working days to deliver an order.

- Calculate annual demand.
- Determine average daily demand.
- Calculate the reorder point.
- Explain what the calculated reorder point means for the company.

Question 10: Precision Tools Ltd. uses 9,600 industrial bearings annually. The cost of placing an order is Tk. 125 per order, while the annual holding cost is Tk. 8 per bearing. The company operates 288 working days per year. Supplier lead time is 6 working days. Determine the following:

- Economic Order Quantity.
- Number of orders placed during a year.
- Length of the order cycle in working days.
- Total annual ordering and holding cost at EOQ.
- Reorder point.
- Draw approximately two inventory cycles and clearly show the reorder point and lead time.

Question 11: PureCap Industries Ltd. uses 36,000 plastic bottle caps annually. Each replenishment order costs Tk. 900. The annual holding cost is Tk. 18 per unit. The company currently orders 3,000 units each time.

- Calculate annual ordering cost under the current policy.
- Calculate annual holding cost under the current policy.
- Calculate the current total annual inventory cost.

- d) Determine the EOQ.
- e) Calculate total annual inventory cost at EOQ.
- f) Determine the annual savings if the company changes from its current policy to the EOQ policy.

Question 12: Secure Tech Retail Ltd. sells 6,000 smart access control devices per year. Each device costs Tk. 1,200. The cost of placing an order is Tk. 1,500. Annual holding cost is estimated at 15 percent of the purchase value per unit. The business operates 300 working days per year.

- a) Determine the annual holding cost per unit.
- b) Calculate the EOQ.
- c) Calculate annual ordering cost and annual holding cost at EOQ.
- d) Determine total annual inventory cost.
- e) Determine the number of orders per year and the approximate length of the order cycle in working days.

Question 13: Bright Lighting Ltd. purchases LED driver modules from a supplier. Annual demand is 18,000 units. Ordering cost is Tk. 300 per order and annual holding cost is Tk. 10 per unit. The normal purchase price is Tk. 50 per unit. The supplier offers a 6 percent discount if the company orders 2,000 units at a time.

- a) Calculate the EOQ using the normal inventory information.
- b) Calculate total annual cost at EOQ, including purchase cost, ordering cost, and holding cost.
- c) Determine the discounted unit price.
- d) Calculate total annual cost if the company orders 2,000 units each time and receives the discount.
- e) Should the company accept the supplier's offer? Support your answer with calculations.

Question 14: Metro Packaging Ltd. imports food grade aluminum containers for one of its production lines. Average demand is 2,500 units per month. The company operates 48 working weeks per year. The following information is available:

- Ordering cost is Tk. 2,400 per order.
- Annual holding cost is Tk. 30 per unit.
- Normal purchase price is Tk. 500 per unit.
- Current order quantity is 2,500 units.
- Supplier lead time is 3 working weeks.
- The supplier offers a 2 percent discount if 5,000 units are purchased in one order.

- a) Determine annual demand and EOQ.
- b) Calculate annual inventory cost under the current ordering policy.
- c) Calculate annual inventory cost at EOQ.
- d) Determine the reorder point.
- e) Calculate total annual cost at EOQ, including purchase cost.
- f) Calculate total annual cost if the company accepts the discount and orders 5,000 units at a time.
- g) Recommend the better purchasing policy.
- h) If the recommended policy is followed for two years, calculate the total cost advantage compared with the EOQ policy.

Question 15: Urban Apparel Ltd. uses an imported zipper for garment production. Average requirement is 1,200 boxes per month. Each order placed with the supplier costs Tk. 3,600. Annual holding cost is Tk. 48 per box.

The Purchasing Department is considering the following two policies:

- Policy A: Order 1,200 boxes once each month.
- Policy B: Order 600 boxes twice each month.

- a) Calculate annual demand.
- b) Calculate annual ordering cost, annual holding cost, and total annual inventory cost under Policy A.
- c) Calculate the same costs under Policy B.
- d) Determine which of the two policies has the lower annual inventory cost.
- e) Calculate the EOQ.
- f) Calculate total annual inventory cost at EOQ.
- g) Compare the EOQ policy with the better of the two existing policies and determine the additional annual savings.

Question 16: Solar Grid Solutions Ltd. is installing a rooftop solar power system for an industrial facility.

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
A	Conduct site survey	None	4
B	Review electrical load	None	6
C	Assess roof structure	A	5
D	Design panel layout	A	7
E	Select inverter system	B	3
F	Prepare procurement plan	C, E	6
G	Prepare rooftop area	D	5
H	Install solar equipment	F, G	8
I	Complete permit documents	E	3
J	Conduct final commissioning	H, I	4

- a) Draw the project network diagram.
- b) Determine the critical path and project completion time.
- c) Calculate ES, EF, LS, LF, and slack for all activities.
- d) If Activity F is delayed by 1 week, determine whether the project completion time will change.

Question 17: Cold Chain Services Ltd. is developing a temperature controlled distribution facility.

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
A	Clear construction site	None	5
B	Specify refrigeration system	None	4
C	Arrange utility connection	None	6
D	Complete foundation work	A	7
E	Secure building approval	A	3
F	Source refrigeration equipment	B	9

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
G	Install power substation	C	5
H	Complete structural erection	D, F	6
I	Obtain compliance clearance	E, G	8
J	Install insulation system	H	5
K	Test refrigeration system	I	7
L	Complete facility handover	J, K	4

- Draw the network diagram and determine the project completion time.
- Identify the critical path.
- Determine the second longest path. How much delay can this path absorb without affecting the project completion time?
- Because of a temporary resource shortage, either Activity F or Activity G must be delayed by 2 weeks. Which activity should management delay to reduce the impact on the project? Explain.
- If the duration of Activity K is reduced from 7 weeks to 4 weeks, determine the revised project completion time.

Question 18: Dairy Pro Foods Ltd. is installing a new automated dairy packaging line.

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
A	Define packaging requirements	None	4
B	Prepare utilities plan	None	5
C	Review regulatory requirements	None	3
D	Redesign production layout	A	6
E	Select packaging format	A	5
F	Upgrade plant utilities	B	7
G	Validate hygiene controls	B, C	4
H	Install main equipment	D, F	8
I	Configure quality system	E, G	6
J	Prepare maintenance spares	F	5
K	Calibrate production equipment	H	6
L	Integrate control system	I, J	7
M	Conduct trial production	K, L	5
N	Train shift operators	G	4
O	Complete production handover	M, N	3

- Draw the project network diagram.
- Determine the critical path and project duration.
- Calculate the slack for all activities.
- If Activity G is delayed by 3 weeks, determine the effect on project completion.
- Management is considering reducing Activity F by 3 weeks. Determine the revised project duration and explain why the overall duration may not reduce by the full 3 weeks.

Question 19: Data Core Ltd. is moving a company's existing information systems to a new data center.

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
A	Audit existing servers	None	3
B	Procure new hardware	A	5
C	Configure network design	A	4
D	Prepare data center site	None	6
E	Install server racks	D	4
F	Install hardware units	B, E	3
G	Migrate core applications	C, F	5
H	Perform security testing	G	3
I	Train support team	C	2
J	Complete system go live	H, I	2

- Draw the project network diagram.
- Calculate the earliest start and earliest finish time for all activities.
- Determine the critical path and project completion time.
- Prepare a Gantt chart using the earliest start schedule.
- If Activity B is delayed by 1 week, determine whether the final project completion time will change.

Question 20: Smart Ware Ltd. is implementing an automated warehouse management system.

Activity	Description	Immediate Predecessor(s)	Duration in Weeks
A	Conduct warehouse survey	None	4
B	Design operating process	None	6
C	Prepare vendor shortlist	A	5
D	Redesign warehouse layout	A	7
E	Develop system specification	B	6
F	Evaluate vendor proposals	C, E	4
G	Upgrade utility services	D	5
H	Develop warehouse software	E	8
I	Order automation equipment	F	6
J	Complete foundation works	G	4
K	Install automation equipment	I, J	5
L	Integrate hardware and software	H, K	6
M	Train warehouse users	H	3
N	Conduct pilot go live	L, M	4

- Draw the project network diagram.
- Determine the critical path and overall project duration.
- Calculate the slack of Activities G and H.
- Identify the second longest path and determine how much delay it can absorb without affecting the project completion time.
- Due to limited resources, management must delay either Activity G or Activity H by 4 weeks. Which activity should be delayed? Support your decision using the project schedule.
- Management is considering reducing Activity I from 6 weeks to 3 weeks. Determine the revised project completion time. Explain why reducing the duration of a critical activity by 3 weeks does not necessarily reduce the total project duration by the same amount.

AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Department of Computer Science

Assignment 03

CSC01379: Engineering Management

Semester: [Summer] | Academic Year: 2025-2026

Section: B

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Answer to the Q. No - 01

a) True

b) True

c) True

d) False. $\Rightarrow$ Safety stock is maintained to reduce the risk of stockout caused by uncertainty in demand and LT.

e) True

f) True

g) True

h) False $\Rightarrow$ The least cost method provides a basic feasible solution but it does not always guarantee the minimum possible transportation cost.

i) True

j) False $\Rightarrow$ Reduction within its available slack normally does not reduce project duration.

Answer to the Q. No - 02

a) C. Financial flow

b) b. Sourcing

c) b. Dummy destination of 60 units.

d) C. Periodic review system

e) C. TK. 100

f) a. Increase

g) C. 200 units

h) e. Not Monitoring and Controlling

i) b. Lowest LS among successors.

j) b. Gantt chart.

Answer to the Q. No-03

a) Here,
the main driver is Facilities
and primary objective is Responsiveness.

The company opens several fulfilled facilities close to customers. This reduces delivery time and improves responsiveness, although facility related expenditure increases.

b) Here,
the main driver is Information
and primary objective is Responsiveness.

Real-time inventory, sales and shipment information allows the company to identify potential stockouts and transfer inventory before customers are affected. It may also improve efficiency because inventory can be shared across depots, but the immediate objective is responsiveness.

c) Here,
the main driver is Inventory
and primary objective is Efficiency

The company keeps low inventory and a limited product range to reduce storage cost, handling cost, obsolescence and capital tied up in inventory.

d) Here,

The main driver is Sourcing
and primary objective is Efficiency.

The firm outsources repair and revenue-logistics activities instead of maintaining internal resources. The decision reduces internal fixed and operating costs.

e) Manufacturing the same appliance every working day is an operation because it is repetitive and continuous.

Installing a new robotic assembly line is a project because it is unique and has defined deliverables, budget, starting point and six month target.

f) The pharmacy uses a periodic review system because inventory is checked at a fixed interval, every Friday, and the order quantity is determined at the review date.

g) Here,

$$\begin{aligned}\text{Shortage} &= \text{Total Demand} - \text{Total Supply} \\ &= 365 - 340 \\ &= 25\end{aligned}$$

A dummy source with a supply of 25 units and zero transportation costs should be added. The dummy allocation represents unmet demand or the amount that can not be supplied by the actual sources.

b) A Gantt chart is most appropriate. It shows planned activity start and finish times and allows actual progress to be compared with planned progress.

_____ x _____

Answer to the Q. No - 04

a) Hence,

$$\text{Total Supply, } S_T = 70 + 60 + 50 \\ = 180$$

$$\text{Total Demand, } D_T = 80 + 55 + 45 \\ = 180$$

$$\therefore S_T = D_T$$

So, The problem is balanced.

b)

Source	Dhaka	Raj	Rhulna	Supply
Grozipur	5 (70)	8	8	70
Nanayagarj	6 (5)	2 (55)	8	60
Cumila	4 (5)	7	1 (45)	50
Demand	80	55	45	180/180

c) Hence,

$$TC = (1 \times 45) + (2 \times 55) + (4 \times 5) + (5 \times 70) + (6 \times 5) \\ = 555$$

_____ x _____

Answer to the Q. No - 05

a) Here,

$$\begin{aligned}\text{Total Supply} &= 55 + 45 + 60 + 40 \\ &= 200\end{aligned}$$

$$\begin{aligned}\text{Total Demand} &= 80 + 70 + 65 \\ &= 215\end{aligned}$$

$\therefore$ Total Supply $\neq$ Total Demand

So, The problem is unbalanced.

b) Add a dummy source of $(215 - 200)$ or 15 units with zero cost.

c)

Source	Hos - A	Hos - B	Hos - C	Supply
Depot 1	9 (20)	4 (30)	11 (5)	55
Depot 2	6	12	3 (45)	45
Depot 3	5 (60)	8	10	60
Depot 4	7	2 (40)	13	40
D ₁	0	0	0 (15)	
Demand	80	70	65	215/200

Now,

$$\begin{aligned}\text{TC} &= (2 \times 40) + (3 \times 45) + (4 \times 80) + (5 \times 60) + (9 \times 20) + (11 \times 5) + (0 \times 15) \\ &= 870 \quad \text{Ans.}\end{aligned}$$

d) The 15-unit dummy allocation represents hospital demand that can not be supplied from the four actual depots. It is balancing entry, not an actual shipment.

Answer to the Q. No - 06

a) Here,

$$\begin{aligned}\text{Total Supply} &= 90 + 75 + 65 \\ &= 230\end{aligned}$$

$$\begin{aligned}\text{Total Demand} &= 55 + 60 + 50 + 45 \\ &= 210\end{aligned}$$

$$\begin{aligned}\therefore \text{Difference} &= 230 - 210 \\ &= 20\end{aligned}$$

b) Add a dummy destination with demand of 20 units and zero cost.

c)

Source	North DC	South DC	East DC	West DC	D_1	Supply
Fa-A	7 (10)	3 (60)	11	5	0 (20)	90
Fa-B	9 (25)	6	2 (50)	12	0	75
Fa-C	4 (20)	10	8	1 (45)	0	65
Demand	55	60	50	45		210/230

d) Now,

$$\begin{aligned}TC &= (0 \times 20) + (1 \times 45) + (2 \times 50) + (3 \times 60) + (4 \times 20) + (7 \times 10) + (9 \times 25) \\ &= 700\end{aligned}$$

e) The dummy allocation represents 20 units of unused production capacity. The units are not physically transported to a destination.

Answer to the Q. No - 07

a) Here,

$$\begin{aligned}\text{Total supply} &= 75 + 65 + 55 + 45 \\ &= 240\end{aligned}$$

$$\begin{aligned}\text{Total Demand} &= 60 + 50 + 70 + 60 \\ &= 240\end{aligned}$$

$$\therefore \text{Total Supply} = \text{Total Demand}$$

So, the problem is balanced.

b)

Source	Market 1	Market 2	Market 3	Market 4	Supply
Center A	8 (15)	3 (50)	11 (5)	14 (5)	75
Center B	5	12	2 (65)	9	65
Center C	15	6	10	1 (55)	55
Center D	4 (45)	13	7	16	45
Demand	60	50	70	60	240/240

c) Here,

$$\begin{aligned}\text{TC} &= (1 \times 55) + (2 \times 65) + (3 \times 50) + (4 \times 45) + (8 \times 15) + (11 \times 5) + (14 \times 5) \\ &= 760\end{aligned}$$

d) The largest allocation is Center B to Market 3 = 65 units

_____ x _____

Answer to the Q. No - 08

a) Here,
$$\text{Total Supply} = 120 + 100 + 90 + 80 + 70$$
$$= 460$$

$$\text{Total Demand} = 85 + 95 + 100 + 90 + 70$$
$$= 440$$

So, the problem is unbalanced.

b) Add a dummy hub with demand of $(460 - 440)$ or 20 units and zero transportation costs.

c)

Source	Hub A	Hub B	Hub C	Hub D	Hub E	D_1	Supply
Plant 1	14	7 (15)	18	4 (20)	22 (65)	0 (20)	120
Plant 2	9	16	3 (100)	20	11	0	100
Plant 3	5 (85)	24	13	8	19 (5)	0	90
Plant 4	17	2 (80)	21	10	15	0	80
Plant 5	6	12	25	1 (70)	23	0	70
Demand	85	95	100	90	70		440/460

d) $TC = 0 \times 20 + 1 \times 70 + 2 \times 80 + 3 \times 100 + 4 \times 20 + 5 \times 85 + 7 \times 15 + 19 \times 5 + 22 \times 65$

$$= 2665 \quad \text{Ans.}$$

e) The dummy allocation represents 20 units of production capacity that remain unused because total capacity exceeds total hub demand.

Answer to the Q. No - 09

a) Here,

$$\begin{aligned}\text{Annual demand} &= 2400 \times 12 \\ &= 28800\end{aligned}$$

$$\therefore D = 28800 \text{ Ans.}$$

b) We know,

$$\begin{aligned}\text{Daily Demand, } d &= \frac{D}{\text{Store open}} \\ &= \frac{28800}{288}\end{aligned}$$

$$= 100 \text{ Ans.}$$

Here,

$$\text{Store open} = 288$$

c) We know,

$$ROP = d \times LT$$

$$= 100 \times 6$$

$$= 600$$

Ans.

Here,

$$LT = 6$$

d) When inventory falls to 600 cartridges, the company should place a new order. The 600 units are expected to satisfy demand during the six-day lead time.

_____ x _____

Answer to the Q. No-10

a) we know,

$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2DS}{H}}$$

Here,

$$D = 9600$$

$$S = 125$$

$$H = 8$$

$$= \sqrt{\frac{2 \times 9600 \times 125}{8}}$$

$$= 547.72 \approx 548$$

Ans.

b) we know,

$$\text{Number of orders, } N = \frac{D}{EOQ}$$

$$= \frac{9600}{547.72}$$

$$= 17.53 \text{ Ans.}$$

$$\text{c) Order-cycle length} = \frac{EOQ}{D} \times \text{store open}$$

$$= \frac{547.72}{9600} \times 288$$

$$= 16.43 \approx 17 \text{ working days.}$$

Ans.

d) we know,

$$TC = \frac{EOQ}{2} H + \frac{D}{EOQ} S$$

$$= \frac{547.72}{2} \times 8 + \frac{9600}{547.72} \times 125$$

$$= 2190.89 + 2190.9$$

$$= 4381.78 \text{ Ans.}$$

e) we know,

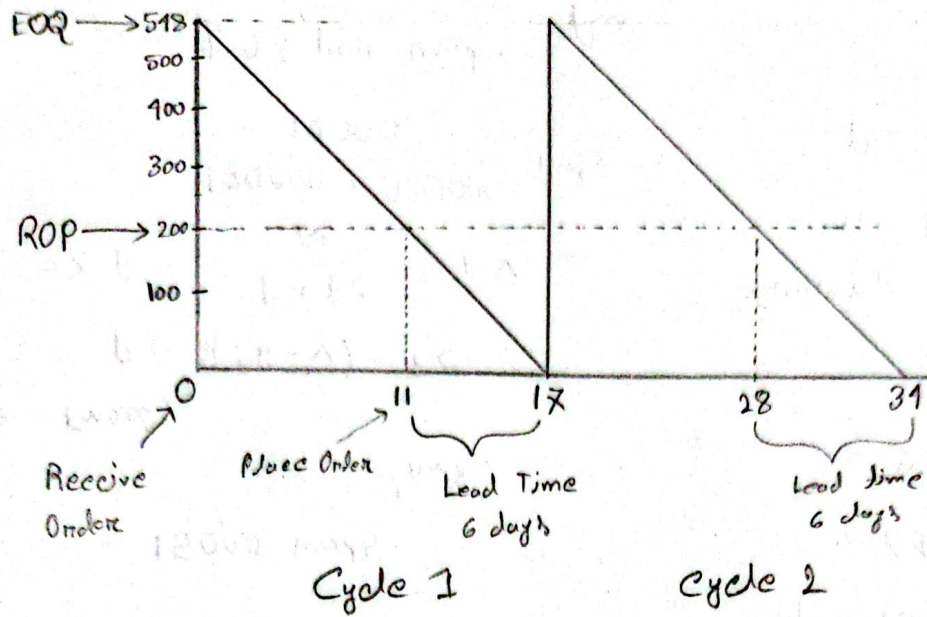
$$ROP = \alpha \times LT$$

$$= \frac{9600}{278} \times 6$$

$$= 33.33 \times 6$$

$$= 200 \text{ Ans.}$$

b)



Answer to the Q. No - 11

a) Ordering Cost, $AOC = \frac{D}{Q} \times S$

$$= \frac{3600}{3000} \times 900$$

$$= 10800 \text{ Ans.}$$

Given,

$$D = 3600$$

$$S = 900$$

$$H = 18$$

$$Q = 3000$$

b) Holding Cost, $AHC = \frac{Q}{2} \times H$

$$= \frac{3000}{2} \times 18$$

$$= 27000 \text{ Ans.}$$

c) $TC = AOC + AHC$

$$= 10800 + 27000$$

$$= 37800 \text{ Ans.}$$

d) $EOQ = \sqrt{\frac{2DS}{H}}$

$$= \sqrt{\frac{2 \times 3600 \times 900}{18}}$$

$$= 1897.37 \text{ units.}$$

Ans.

e) Cost at EOQ,

$$\begin{aligned} TC &= \frac{EOQ}{2} H + \frac{D}{EOQ} S \\ &= \frac{1897.37}{2} \times 18 + \frac{36000}{1897.37} \times 900 \\ &= 34152.60 \text{ Am.} \end{aligned}$$

f) Hence,

$$\begin{aligned} \text{Annual Savings} &= 37800 - 34152.60 \\ &= 3647.4 \text{ Ans.} \end{aligned}$$

Answer to the Q. No - 12

a) Here,

$$\begin{aligned} \text{Annual holding Cost, } H &= 1200 \times 15\% \\ &= 180 \text{ per device/year} \end{aligned}$$

b) Given,

$$D = 6000$$

$$S = 1500$$

$$H = 180$$

we know,

$$\begin{aligned} EOQ &= \sqrt{\frac{2DS}{H}} \\ &= \sqrt{\frac{2 \times 6000 \times 1500}{180}} \\ &= 316.23 \text{ devices.} \end{aligned}$$

c) Here,

$$\begin{aligned} \text{Annual Ordering Cost, AOC} &= \frac{D}{EOQ} \times S = \frac{6000}{316.23} \times 1500 \\ &= 28460.3 \text{ Am.} \end{aligned}$$

$$\text{Annual Holding Cost, AHC} = \frac{EOQ}{2} \times H$$

$$= \frac{316.23}{2} \times 180 = 28460.7 \text{ Am.}$$

d) Now,

$$TC = AOC + AHC$$

$$= 28460 \cdot 3 + 28460 \cdot 7$$

$$= 56921 \quad \text{Ans.}$$

e) Here,

$$\text{Number of orders, } N = \frac{D}{EOQ} = \frac{6000}{316.23} \text{ order/year}$$

$$= 18.97 \text{ order/year} \quad \text{Ans.}$$

$$\text{and Cycle, } = \frac{EOQ}{D} \times \text{days open.}$$

$$= \frac{316.23}{6000} \times 300$$

$$= 15.81 \text{ working days.}$$

Ans.

Answer to the Q. No - 13

a) we know,

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$= \sqrt{\frac{2 \times 18000 \times 300}{10}}$$

$$= 1039.23 \text{ units.}$$

Given,

$$D = 18000$$

$$S = 300$$

$$H = 10$$

$$P = 50$$

Ans.

b) Using EOQ,

$$\text{Purchase Cost} = DP$$

$$= 18000 \times 50$$

$$= 900000$$

Ans.

$$\text{Ordering Cost} = \frac{D}{EOQ} \times S$$

$$= \frac{18000}{1039.23} \times 300$$

$$= 5196.15$$

Ans.

$$\begin{aligned}\text{and Holding Cost} &= \frac{EOQ}{2} H \\ &= \frac{1039.23}{2} \times 10 \\ &= 5196.15 \text{ Ans.}\end{aligned}$$

$$\begin{aligned}\text{Now,} \\ TAC &= 5196.15 + 5196.15 + 900000 \\ &= 910392.3\end{aligned}$$

Ans.

c) Now,

$$\begin{aligned}\text{Discounted Price, } P_d &= 50 (1 - 0.06) \\ &= 47\end{aligned}$$

d) Cost at $Q = 2,000 \Rightarrow$

$$\begin{aligned}\text{Total Annual Cost, } TAC_d &= \frac{Q}{2} H + \frac{D}{Q} \times S + DP_d \\ &= \frac{2000}{2} \times 10 + \frac{18000}{2000} \times 300 + 18000 \times 47 \\ &= 858700 \text{ Ans.}\end{aligned}$$

e) Now,

$$\begin{aligned}\text{Savings} &= 910392.3 - 858700 \\ &= 51692.3\end{aligned}$$

The company should accept the supplier's discount offer because the discounted total annual cost is lower.

_____ x _____

Answer to the Q. No - 14

a) Here,

$$\text{The annual Demand, } D = 2500 \times 12 \\ = 30,000$$

Given,

$$S = 2400$$

$$H = 30$$

Now,

$$EOQ = \sqrt{\frac{2DS}{H}} \\ = \sqrt{\frac{2 \times 30000 \times 2400}{30}} \\ = 2190.89 \text{ units.}$$

b) At $Q = 2500$,

$$TC_Q = \frac{Q}{2} \times H + \frac{D}{Q} \times S \\ = \frac{2500}{2} \times 30 + \frac{30000}{2500} \times 2400 \\ = 66300 \text{ Ans.}$$

c) At $EOQ = 2190.89$,

$$TC_{EOQ} = \frac{EOQ}{2} \times H + \frac{D}{EOQ} \times S \\ = \frac{2190.89}{2} \times 30 + \frac{30000}{2190.89} \times 2400 \\ = 65726.71$$

Ans.

d) we know,

$$ROP = 2 \times LT$$

Given,

$$LT = 3$$

$$= \frac{30000}{18} \times 3$$

$$\text{Store open} = 48$$

$$= 1875 \text{ units.}$$

Ans.

e) Total Annual Cost at EOQ,

$$TAC_{EOQ} = DP + TC_{EOQ}$$

$$= 30000 \times 500 + 65726.71$$

$$= 15065726.71$$

Ans.

f) At $Q = 5000$,

$$P_d = 500(1 - 0.02)$$

$$= 490.$$

Now,

$$TAC_d = \frac{Q}{2} H + \frac{D}{Q} S + DP_d$$

$$= \frac{5000}{2} \times 15 + \frac{30000}{5000} \times 2400 + 30000 \times 490$$

$$= 14789400$$

Ans.

g) Here,

$$\text{After discount the savings} = 15065726.71 - 14789400$$

$$= 276326.71$$

So, the discount policy should be accepted.

h) Now,

$$\text{Two year cost advantage} = 2 \times 276326.71$$

$$= 552653.42$$

Ans.

Answer to the Q. No - 15

a) Annual Demand, $D = 1200 \times 12$
 $= 14400$ boxes. Ans.

b) we know,

$$\begin{aligned}\text{Annual Ordering Cost, } AOC &= \frac{D}{Q} S \\ &= \frac{14400}{1200} \times 3600 \\ &= 43200 \text{ Ans.}\end{aligned}$$

Given for Policy A,

$$Q = 1200$$

$$S = 3600$$

$$H = 48$$

$$\begin{aligned}\text{Annual Holding Cost, } AHC &= \frac{Q \times H}{2} \\ &= \frac{1200 \times 48}{2} \\ &= 28800\end{aligned}$$

Ans.

$$\begin{aligned}TC_A &= AOC + AHC \\ &= 43200 + 28800 \\ &= 72000 \text{ Ans.}\end{aligned}$$

c) we know,

$$\begin{aligned}AOC &= \frac{D \times S}{Q} \\ &= \frac{14400}{600} \times 3600 \\ &= 86400\end{aligned}$$

Given for Policy B,

$$Q = 600$$

$$S = 3600$$

$$H = 48$$

$$AHC = \frac{600}{2} \times 48 = 14400$$

$$\begin{aligned}\therefore TC_B &= AOC + AHC = 86400 + 14400 \\ &= 100800 \text{ Ans.}\end{aligned}$$

d) Here we can find,

$$TC_A = 72000$$

$$TC_B = 100800$$

$$\therefore TC_A < TC_B$$

So, Policy A is better.

e) we know,

$$EOQ = \sqrt{\frac{2DS}{H}}$$

$$= \sqrt{\frac{2 \times 14400 \times 3600}{48}}$$

$$= 1469.69 \text{ boxes.}$$

b) at EOQ,

$$AOC = \frac{D}{EOQ} \times S = \frac{14400}{1469.69} \times 3600$$
$$= 35272.65$$

$$AHC = \frac{EOQ}{2} \times H$$
$$= \frac{1469.69}{2} \times 48$$
$$= 35272.65$$

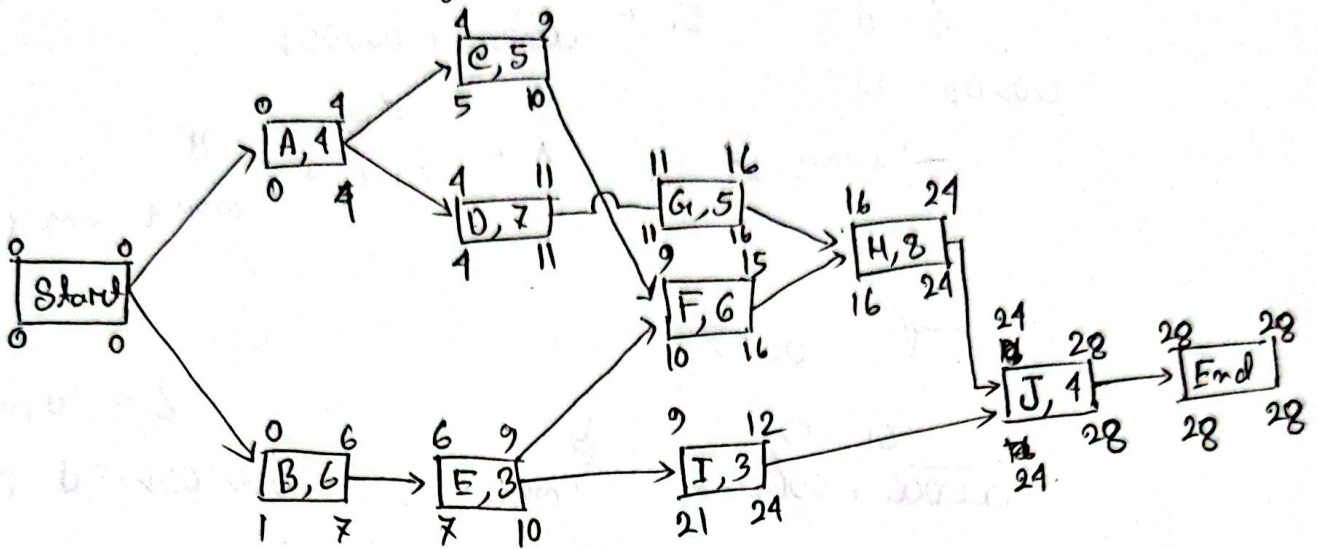
$$TC_{EOQ} = 35272.65 + 35272.65$$
$$= 70545.3 \text{ Rs.}$$

$$f) \text{ Annual Savings} = 72000 - 70545.30$$
$$= 1454.7$$

So, the additional annual saving = 1454.7

Answer to the Q. No - 16

a) Now, Draw the project network diagram;



b) The critical path is: A - D - G - H - J

and project completion time: 28 weeks.

c) we know,

$$\text{Slack} = \text{LS} - \text{ES}$$

Now, Calculate ES, EF, LS, LF and Slack on the table:

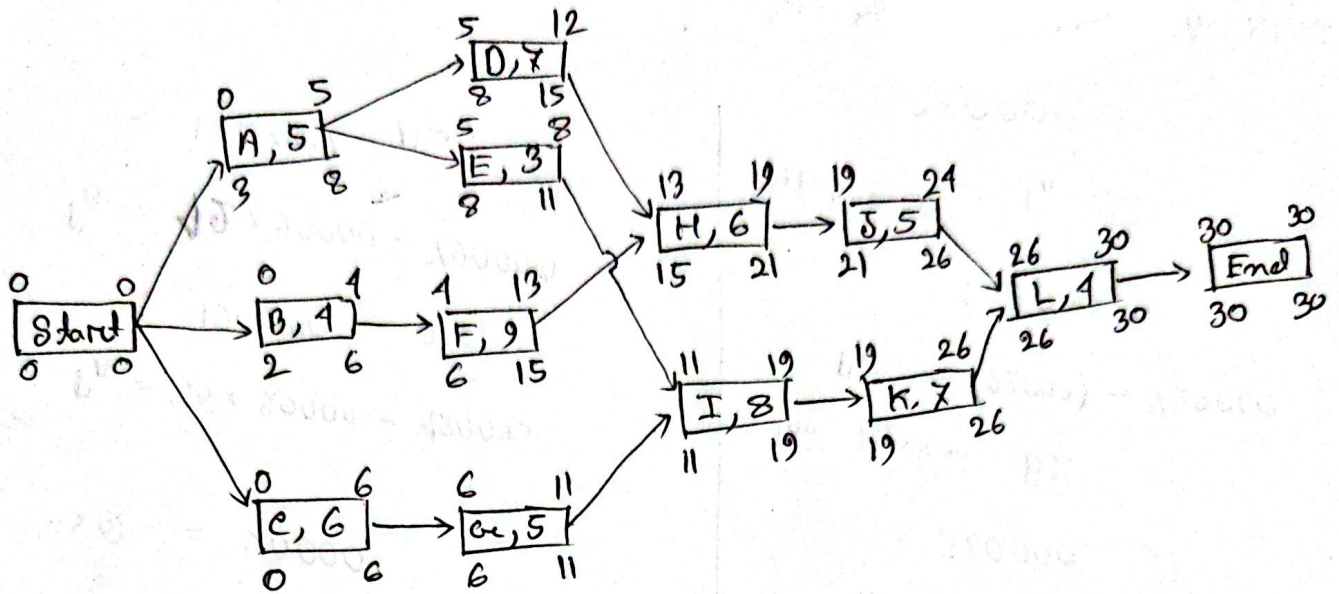
Activity	ES	EF	LS	LF	Slack
A	0	4	0	4	0
B	0	6	1	7	1
C	4	9	5	10	1
D	4	11	4	11	0
E	6	9	7	10	1
F	9	15	10	16	1
G	11	16	11	16	0
H	16	24	16	24	0
I	9	12	21	24	12
J	24	28	24	28	0

d) Here, $\text{Slack}_F = 1 \text{ week}$.

A one-week delay consumes the full slack but does not delay the project completion time (28 weeks). It is beyond week 16. So, the project duration remains 28 weeks.

Answer to the Q. No - 17

a) Now, draw the network diagram:



So, Project completion time = 30 weeks.

b) Critical Path: C - G - I - K - L $\Rightarrow$ 30

c) The second longest path: B - F - H - J - L $\Rightarrow$ 28

So, available delay: $30 - 28 = 2$

$\therefore$ Second longest path slack = 2 weeks

d) Here,

Slack_F = 2 weeks

Slack_G = 0 weeks

Therefore, management should delay activity F by two weeks. This available slack absorbs the delay.

So, Delay F $\Rightarrow$ Project duration remains 30 weeks

$\Rightarrow$ A two week delay in G would increase the duration to 32 weeks.

c) Original critical branch through K:

$$C - G - I - K - L = 30$$

Reduced length:

$$G + 5 + 8 + 4 + 4 = 27$$

However, the other branch through J takes 28 weeks.

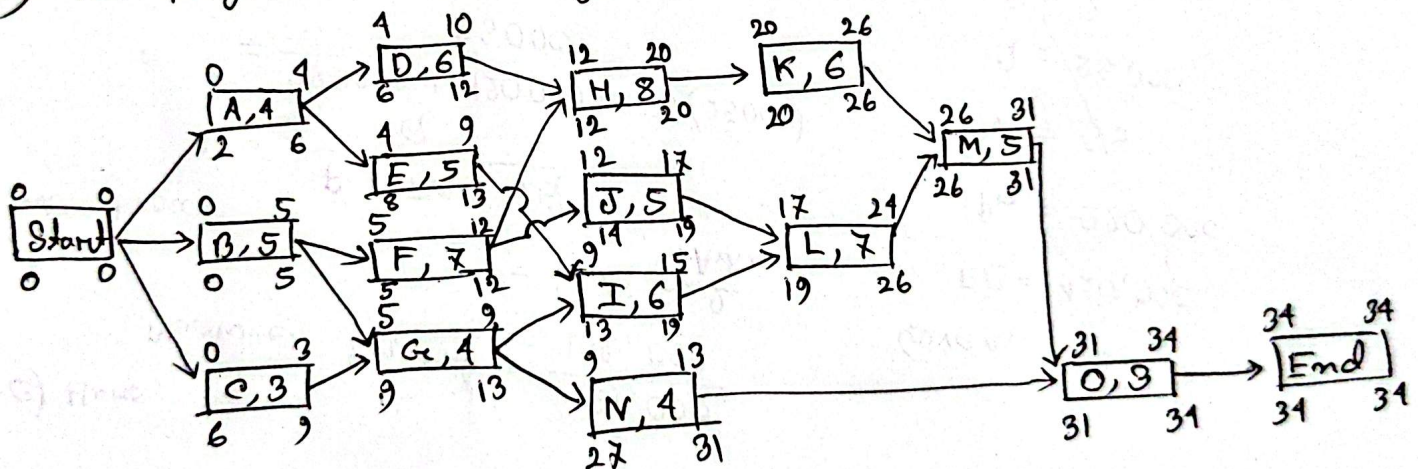
Therefore,

Revised project duration = 28 weeks.

So, the duration decreases by only two weeks because the former second-longest path becomes critical.

Answer to the Q. No - 18

a) The project network diagram:



b) From the network diagram,

The critical path : B - F - H - K - M - O

$$\text{and project duration : } 5 + 7 + 8 + 6 + 5 + 3 = 34$$

c) Hence,

$$\text{Slack}_A = 2 - 0 = 2$$

$$\text{Slack}_E = 8 - 4 = 4$$

$$\text{Slack}_B = 0$$

$$\text{Slack}_F = 0$$

$$\text{Slack}_C = 6 - 0 = 6$$

$$\text{Slack}_G = 9 - 5 = 4$$

$$\text{Slack}_D = 6 - 4 = 2$$

$$\text{Slack}_H = 0$$

$$\text{Slack}_N = 18$$

$$\text{Slack}_I = 4$$

$$\text{Slack}_O = 0$$

$$\text{Slack}_J = 2$$

$$\text{Slack}_K = 0$$

$$\text{Slack}_M = 0$$

d) Here,

Slack of G is 4 weeks.

$$\therefore 3 < \text{Slack}_G$$

So, Project duration remain 34 weeks.

e) F falls from seven weeks to four weeks.

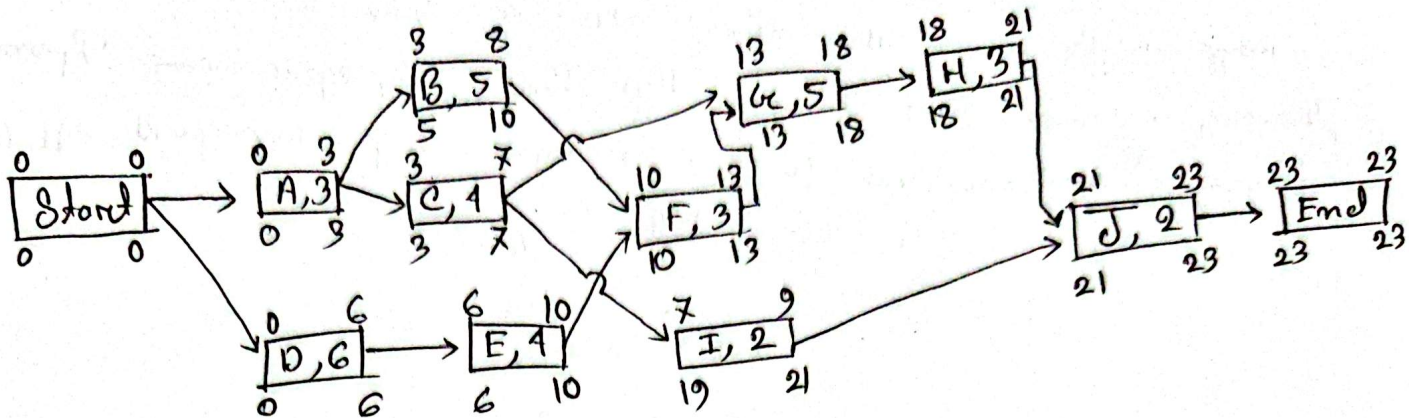
The F branch then allows H to start at week 10 rather than week 12 because O finishes at week 10. Therefore, only two of the three reduced weeks translate into project savings.

So, Revised project duration = 32 weeks.

_____ x _____

Answer to the Q. No - 19

a) The project network diagram:



b)

Activity	ES	EF
A	0	3
B	3	8
C	3	7
D	0	6
E	6	10
F	10	13
G	13	18
H	18	21
I	7	9
J	21	23

c) Here,

the critical path is: D - E - F - G - H - J

and project completion time: 23 weeks.

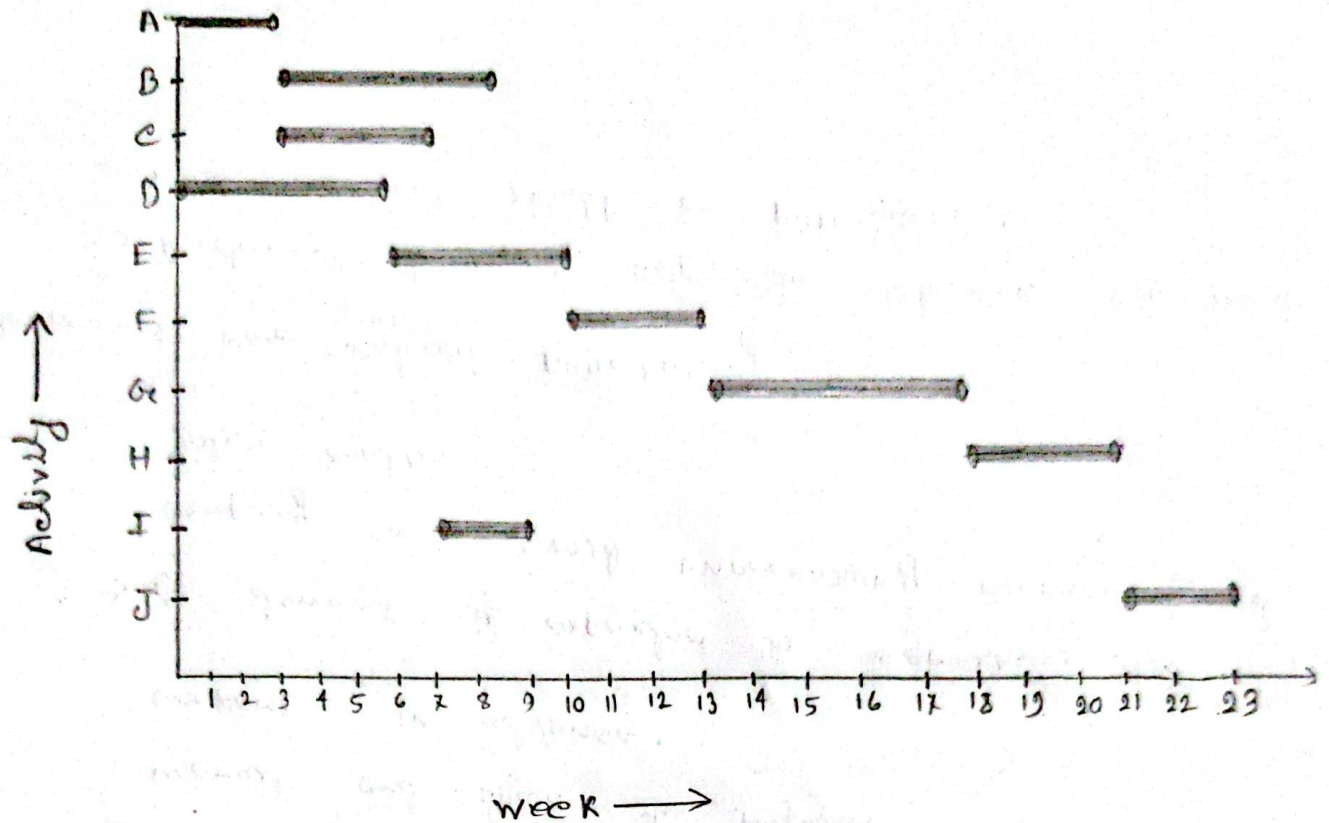
e) For the slack of B is $(5-3)$ or 2 weeks.

Hence, B originally finishes at week 8, while F can not start until E finishes at week 10. A one-week delay means B finishes at week 9. So, F can still start at week 10.

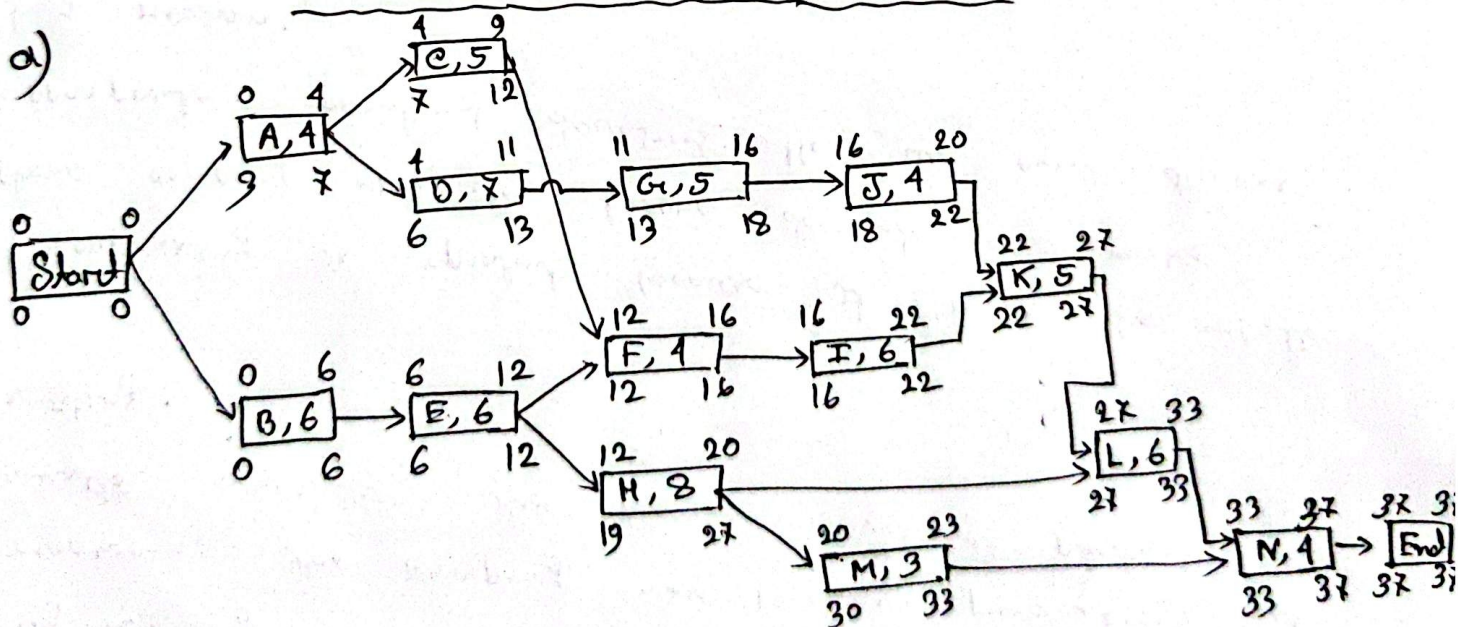
So, the final project duration remains 23 weeks.

P.T.O

d) prepare the Gantt chart using Earliest Start Schedule :



Answer to the Q. No - 20



c) From the 'a' $\Rightarrow$

Slack of G is : $13 - 11 \Rightarrow 2$ weeks.

and Slack of H is : $19 - 12 \Rightarrow 7$ weeks.

d) Here,

The 2nd longest path is : A-D-G-J-K-L-N.

and duration : $4 + 7 + 5 + 4 + 5 + 6 + 4$

$$= 35$$

$$\text{path slack} = 37 - 35$$

$$= 2$$

So, the 2nd longest path has about 2 weeks.

e) we know,

$$\text{Slack}_G = 2 \quad \text{and} \quad \text{Slack}_H = 7$$

So, Delay activity H by four weeks.

The 4 week delay is within H's 7 week slack, so, project duration remains 37 weeks. Delaying G by four weeks would exceed its slack by two weeks and could increase project duration to 39 weeks.

P.T.O

b) Originally,

$$EF_I = 22$$

After reducing I $\Rightarrow EF_I = 19$

However, K depends on both I and J. Activity J finishes at week 20. So, K can not start before week 20.

The revised schedule is:

K: Week 20 to 25

L: Week 25 to 31

M: Week 31 to 35

Therefore, Revised project duration = 35 weeks.

So, the reduction is only two weeks, not three weeks, because Activity J becomes the controlling predecessor of K after I is shortened.
